Minsoo Ha

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Summary

Interested in statistical learning theory, interactive machine learning, and mathematical structure that can be used to make learning systems more reliable and provably correct.

Education

2020 - present B.S. in Mathematics, **POSTECH**, South Korea (GPA: 3.75/4.3, Major 3.9/4.3)

Work Experience

Research Intern, Interactive Machine Learning Lab. at POSTECH Apr 2026 - present

– Research on Machine learning theory, under supervision of Prof. Kwang-Sung Jun

Research Intern, Machine Learning Lab. at POSTECH Sep 2025 - Mar 2026

– Research on LLM reasoning and Reinforcement learning, under supervision of Prof. Sangdon Park

Fundamental Research Engineering Intern, LinqAlpha Sep 2025 - present

– RAG for finance

– Built Text-to-SQL, STT model, and various workflows leveraging LLM

Publications

[1] Junghyun Lee**, Minsoo Ha**, Sanghwa Kim, Yeongjong Kim, Eunjee Lee, Seiyun Shin, and Kwang-Sung Jun. "Pointwise or Pairwise: When Do Pairwise Losses Help Reward Learning, Provably?" In: *arXiv preprint arXiv:2609.37209* (2026).

[2] Chanyeol Choi, Alejandro Lopez-Lira, Yongjae Lee, Jihoon Kwon, Minjae Kim, Juneha Hwang**, Minsoo Ha**, Chaewoon Kim, Jaeseon Ha, Suyeol Yun, et al. "Structuring the unstructured: a multi-agent system for extracting and querying financial kpis and guidance". In: *arXiv preprint arXiv:2505.19197* (2025). Presented at FinIR '25, SIGIR Workshop.

Projects

KnowGem

LLM-generated ontology and knowledge graph construction service for knowledge-integrated reasoning.

Reasoning-meets-lean

Advisor: Prof. Sangdon Park
Integrated the Lean theorem prover into LLM-based mathematical reasoning to improve verifiability.

Structuring the Unstructured financial KPIs and Guidance

Advisor: PhD. Chanyeol Choi
Built full pipeline of extracting financial KPIs and Guidance using LLM, storing in RDB, query by Text-to-SQL.

PPO-scratch Github

Advisor: Prof. Sangdon Park
Proximal Policy Optimization(PPO) algorithm from scratch, compatible with OpenAI Gym API.

Portfolio Risk Optimization via Deep Reinforcement Learning Github

Designed RL setup to optimize risk of S&P 500 portfolio. Trained by PPO algorithm, supports real-time train monitoring.

Financial crisis prediction with Machine Learning

Use Support Vector Machine to classify Financial crisis / Non-crisis / Pre-crisis. Data preprocessing for robust model.

Skills

Mathematics	Real Analysis, Probability Theory
Programming Languages	Python, Typescript, C++, Haskell, Lean, OCaml, SQL
ML Frameworks	Pytorch, Stable-Baselines, Hugging Face Transformers
DevOps / MLOps	Github Actions, Docker, AWS, MLflow, Wandb
Backend Frameworks	FastAPI, Express.js